

Gavin Duong

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Education

University of California, Riverside

Expected in 2026

Bachelor of Science in Computer Science with Business Applications

- **Relevant Coursework:** Applied Machine Learning and Data Mining, Data Management, Edge Computing, Data Structure and Algorithms, Design Operating Systems, Software Engineering, Wireless Networks, Financial Accounting and Management, Marketing and Distribution Channels, Databases for Management

Work Experience

Student Recreation Center at University of California, Riverside

Riverside, CA

Competitive Sports Supervisor

September 2025 - Present

- Supervised 50 Site Managers and Intramural Officials to ensure successful execution of all 4 Intramural Sports and 20+ Clubs and their events.
- Worked directly under the Coordinator and Assistant Director to streamline departmental operations and enhance efficiency.
- Analyzed weekly logistical data to accurately forecast departmental resource needs, regulating the allocation of equipment and facility spaces to optimize operational efficiency.

Site Manager

July 2024 - September 2025

- Supported aid to any technical errors across the departments 12 iPads, 12 radios and two computers.
- Oversaw 30+ Intramural Officials and 200+ participants across all Intramural Sports and Sports Clubs activities such as practices and games.
- Maintained secure and confidential records of incident reports and staff interactions while enforcing strict university recreation policies and safety regulations during all operational hours.

Projects/Certificates

MLB Predictor

March 2026 - Present

Developer

- Engineered a robust data pipeline using Pandas to process over **4,100+ historical MLB player-seasons**, constructing custom composite IDs and calculating complex time-series features while strictly managing missing values.
- Developed a predictive machine learning model with Scikit-Learn (Ridge Regression) to forecast a player's future On-Base Plus Slugging (OPS), achieving a **Mean Absolute Error (MAE) of 0.0683** on out-of-sample data.
- Implemented a custom time-series backtesting framework to simulate real-world forecasting, iteratively training the model across **20+ historical seasons** to prevent data leakage and rigorously validate predictive accuracy.

CookCraft

September 2025 - February 2026

Backend Developer

- Developed a personalized recipe generation application utilizing the **Gemini API**, taking into account the user's inventory, dietary restrictions, and cuisine preferences.
- Utilized **Supabase** as the primary backend service to manage personal preferences, user meal logs, saved recipes, and real-time inventory tracking.
- Implemented secure user sessions using **Supabase Authentication Magic Link**, ensuring that features like profile preferences, recipe creation, and history access are protected by active session tokens.

Generative AI

September 2025

Udemy Certification

- Mastered the implementation of **Large Language Models (LLMs)** using frameworks like **LangChain** and **LlamaIndex** to build context-aware applications.
- Designed and deployed a full-stack **AI Chatbot** integrated with OpenAI's API, managing the end-to-end lifecycle from prompt engineering to API cost optimization.

Target That Muscle

September 2024 - December 2024

Full-Stack Developer

- Developed a C++ terminal-based gym application that utilized **Strategy, Composite, and Abstract Factory** design patterns to enhance code modularity and support scalable feature expansion.
- Ensured **97%+ code reliability** by authoring comprehensive unit tests with **GoogleTest** and conducting memory leak analysis through **Valgrind**.
- Automated build processes using **CMake** and established a **Continuous Integration** pipeline via **GitHub Actions** to streamline deployment and testing.

Technical Skills

Languages: C++, Python, SQL, Node.js (JavaScript, TypeScript)

Frameworks: Next.js, React, Jest

Developer Tools: Supabase, Gemini API, Github, ESLint, Prettier, CMake, Valgrind, ARCGIS, Google Suite